MECHANICS OF RESPIRATION

DR SHAHNEELA SIRAJ
ASSOCIATE PROFESSOR
DIKIOHS, DUHS.

LEARNING OBJECTIVES

At the end of the lecture, students should be able to:

- Define and explain Internal and external respiration
- Define Gas Laws of respiratory physiology
- Describe the role of pressure gradients in causing air flow into and out of lungs
- Explain the role of the respiratory muscles in producing pressure gradients.

MECHANICS OF BREATHING

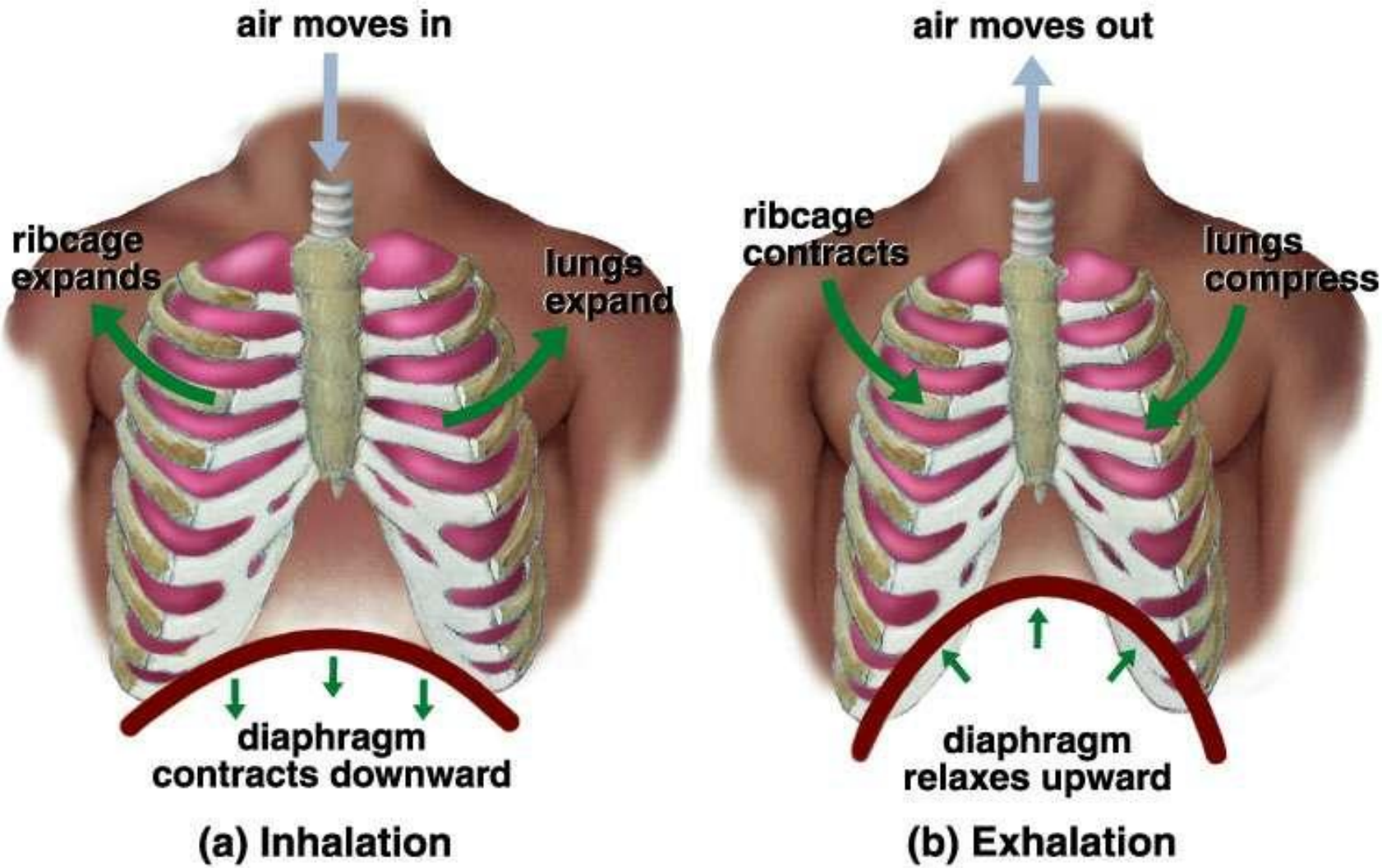
- **Mechanics means science that deals with forces**
- Breathing mechanics is interplay of force generated by pressure, volume and flow changes occurring during the breathing cycle
- Role of respiratory muscles
- Alveolar, Pleural and Transpulmonary pressures
- Compliance of lungs, chest wall & together
- Lung recoil due to elastic & collagen fibers
- Work of breathing.

The three steps of respiration

- **Pulmonary ventilation-** movement of air into and out of lungs
- **Pulmonary respiration** -exchanges of gases between air spaces and blood
- **Tissue respiration** between blood and tissue cells

RESPIRATORY MOVEMENTS

- Respiration occurs in two phases namely inspiration and expiration.
- During inspiration, thoracic cage enlarges and lungs expand so that air enters the lungs easily.
- During expiration, the thoracic cage and lungs decrease in size and attain the pre inspiratory position so that air leaves the lungs easily.
- During normal quiet breathing, inspiration is the active process and expiration is the passive process.


air moves in
ribcage expands
lungs expand
diaphragm contracts downward
(a) Inhalation
air moves out
ribcage contracts
lungs compress
diaphragm relaxes upward
(b) Exhalation

MUSCLES OF RESPIRATION

Muscles of inspiration

- **Principal muscles**
  - Diaphragm
  - External intercostals
- **Accessory muscles**
  - Sternocleidomastoid (lift sternum)
  - Scaleni(elevate first two ribs)
  - Serratus anterior
  - Pectoralis minor (elevate 3rd through 5th ribs)

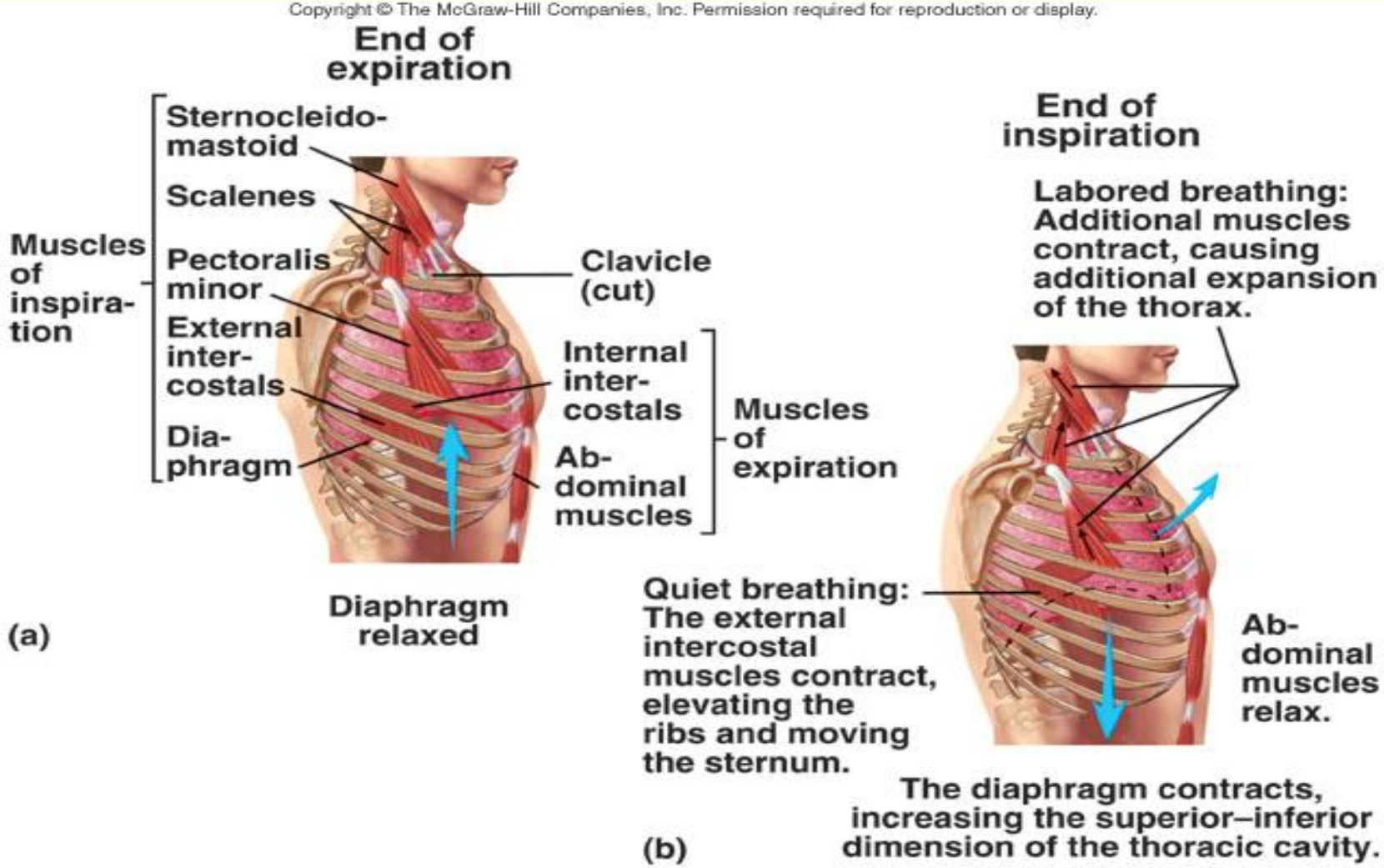
Muscles of normal expiration

- Normal expiration is due to elastic recoil of lungs and associated structures

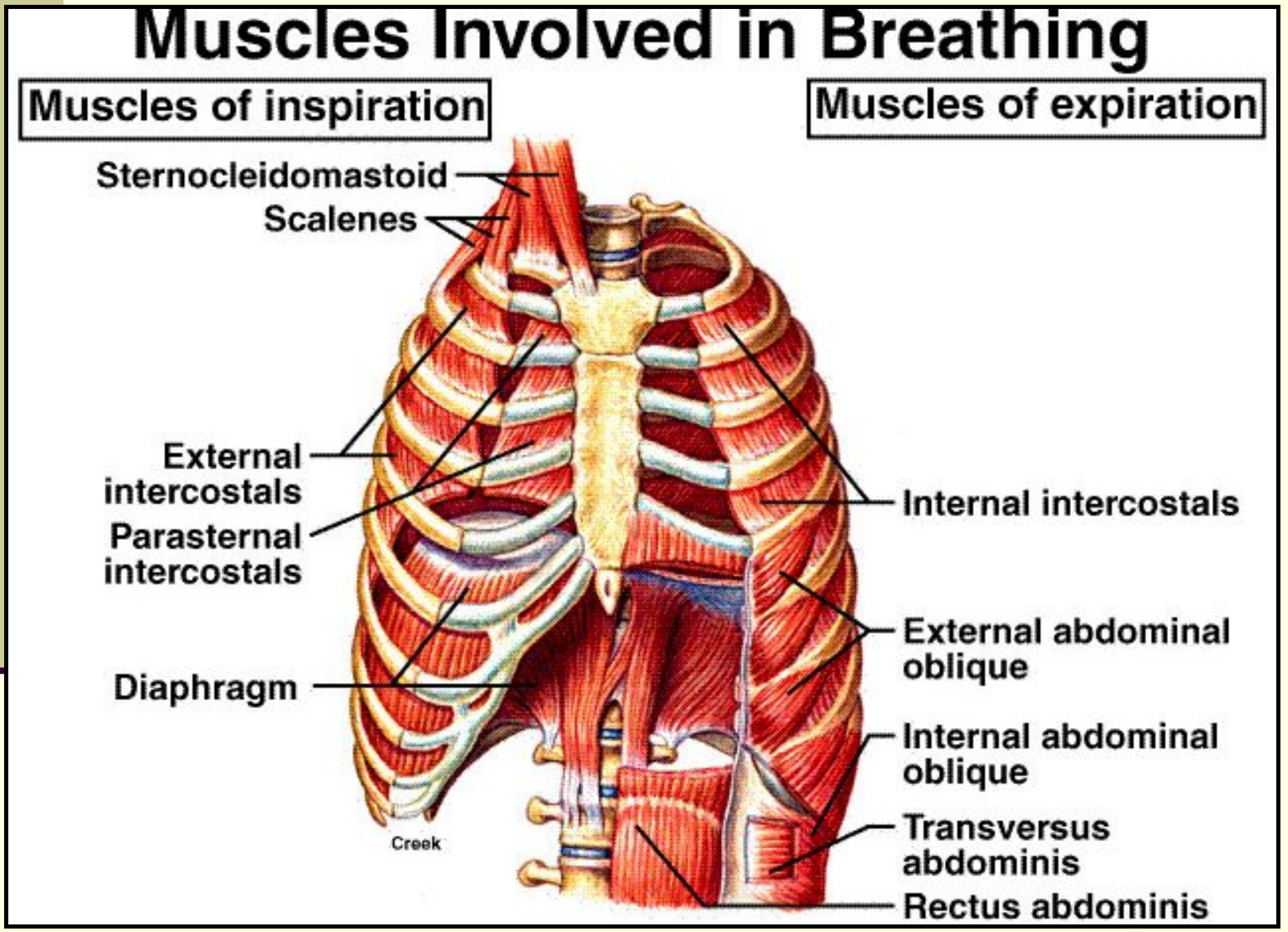
Muscles of forced expiration

- Internal intercostal
- External oblique abdominis
- Internal oblique abdominis
- Rectus abdominis
- Transversus abdominis

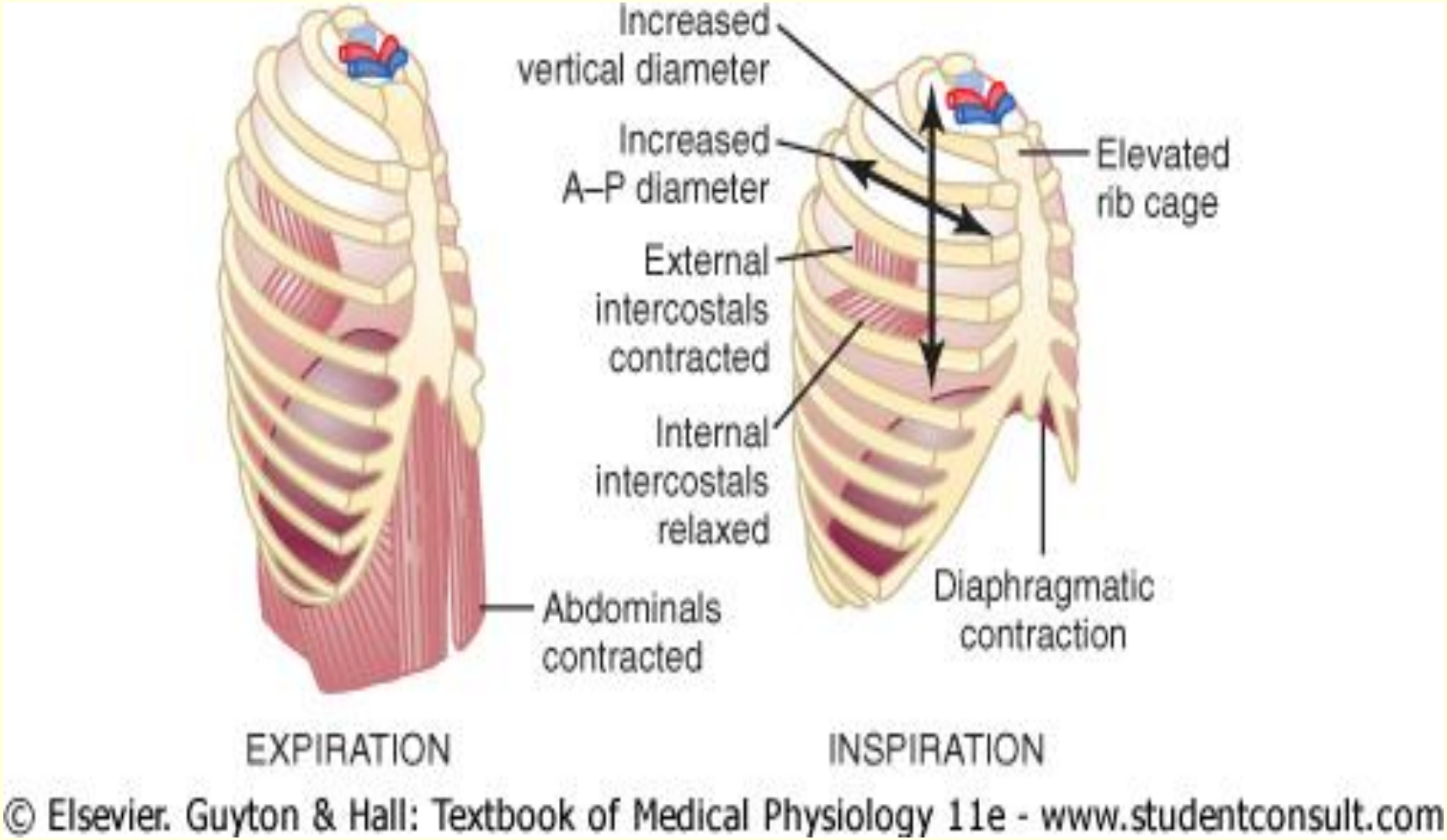
MUSCLES OF RESPIRATION



Muscles Involved in Breathing



FUNCTIONS OF THE RESPIRATORY MUSCLES



© Elsevier. Guyton & Hall: Textbook of Medical Physiology 11e - www.studentconsult.com

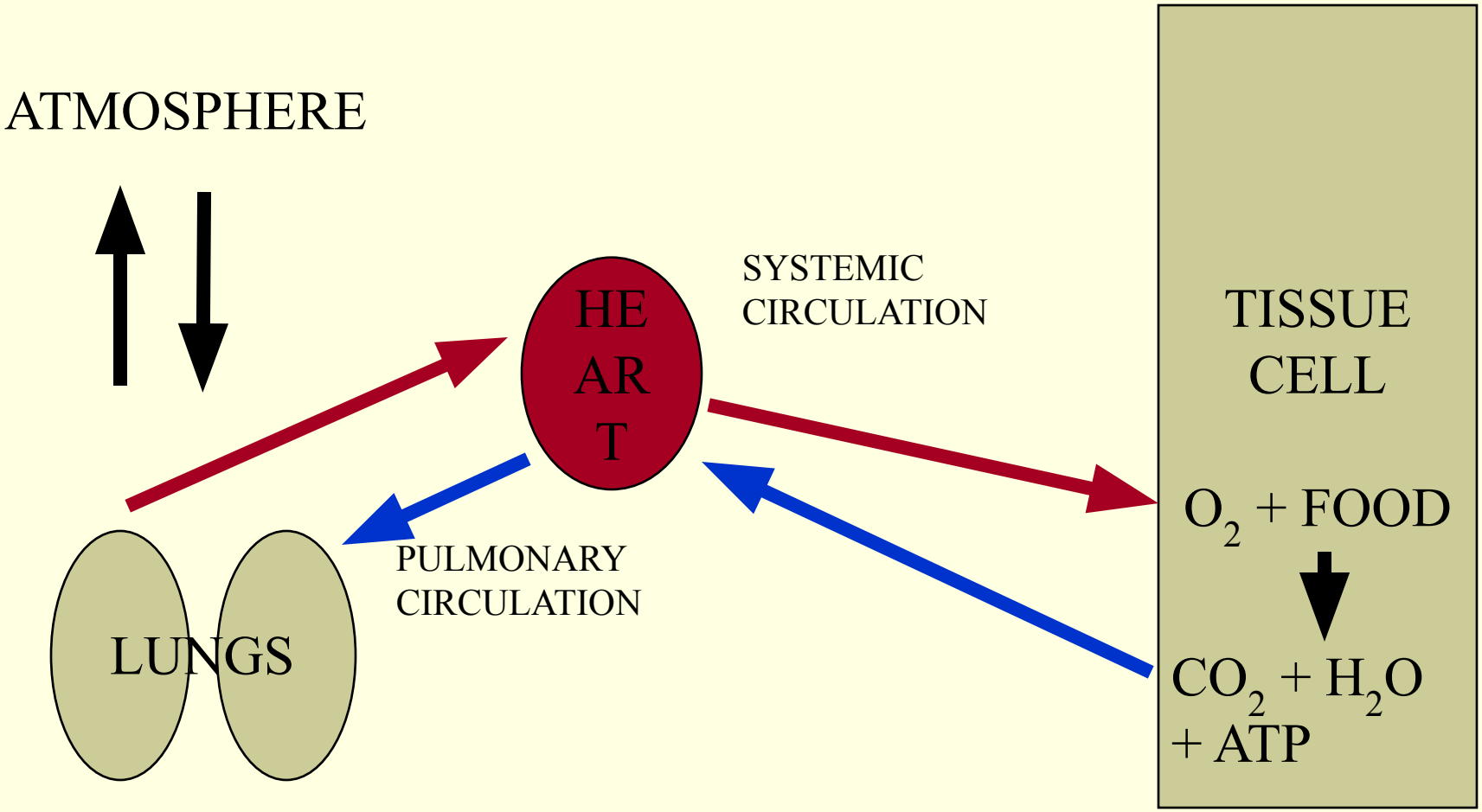
INTERNAL RESPIRATION

- CELLULAR METABOLISM
- ANAEROBIC GLYCOLYSIS
- AEROBIC OXIDATIVE METABOLISM IN THE MITOCHONDRIA

EXTERNAL RESPIRATION

- Ventilation or breathing: air moved in and out of lungs
- Oxygen and Carbon Dioxide exchange in the lungs
- Oxygen and Carbon Dioxide transported by blood to and from tissues
- Exchange of Oxygen and Carbon Dioxide between tissue and blood

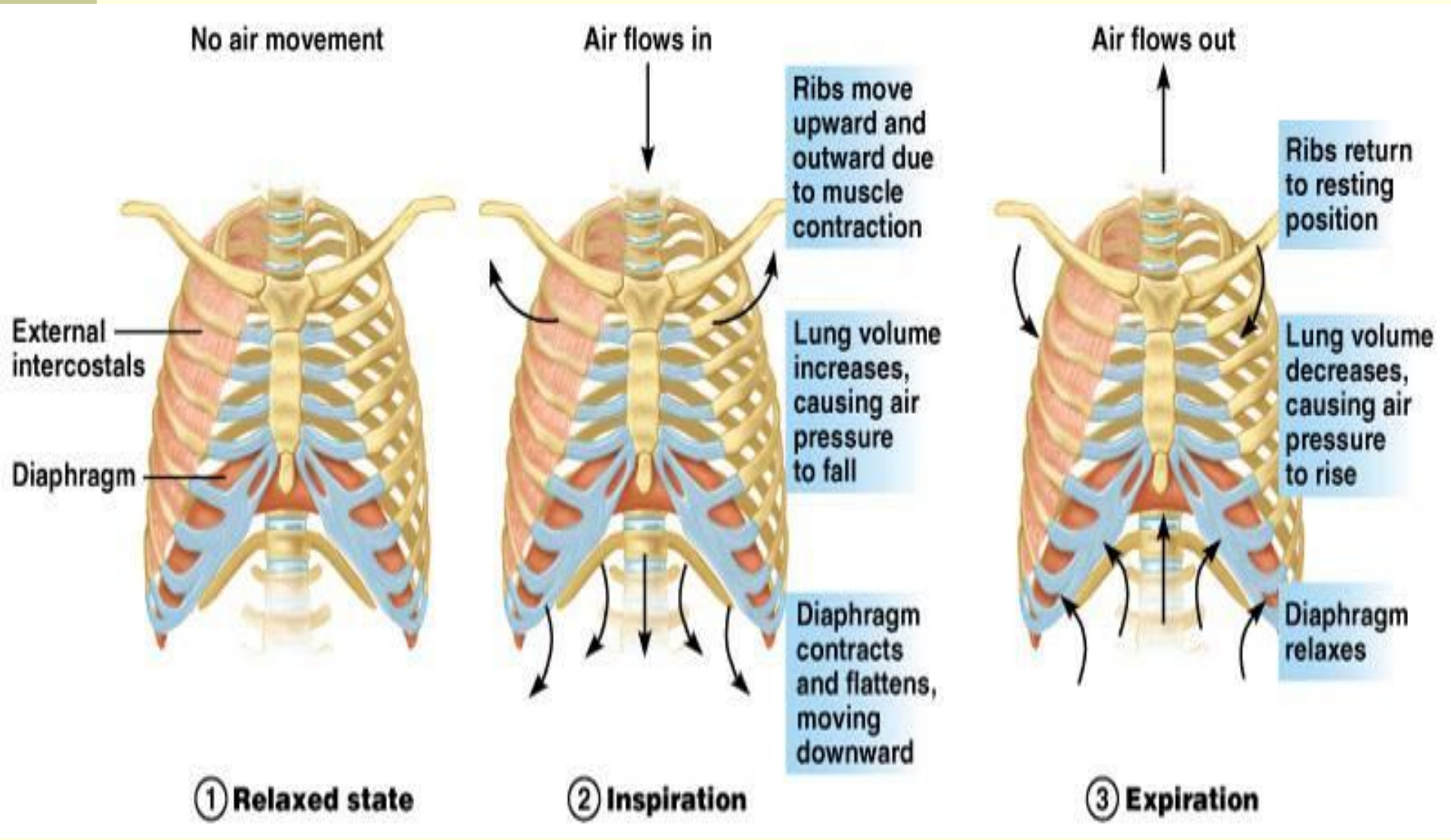
EXTERNAL AND INTERNAL RESPIRATION



Process of Breathing: Pressure Gradient

- Inspiration/Expiration: air in/air out
- Cycle:
  - **Relaxed state:** diaphragm and intercostals muscles relaxed
  - **Inspiration:** diaphragm contracts, pulling muscle down, intercostal muscles contract elevating chest wall and expanding volume of chest, lowering pressure in lungs, pulling in air.
  - **Expiration:** muscles relax, diaphragm resumes dome shape, intercostal muscles allow chest to lower resulting in increase of pressure in chest and expulsion of air.

Respiratory Cycle



- **Resting adult:**

- 10-15 breath/min ; inhales and about 500 ml (during inspiration) and exhales it (during expiration)

PRESSURE AND FLOW

- Air flow governed by same principles of flow, pressure and resistance as blood flow
- Pressure drives respiration – atmospheric (barometric) pressure – the wt of the air above us
- 1 atmosphere (atm) pressure is enough to force a column of mercury 760 mm up an evacuated tube; therefore 1 atm = 760 mm Hg
- This is the average atmospheric pressure at sea level; fluctuates from day to day and is lower at high altitudes

PRESSURE AND FLOW

- If the lungs contain a quantity of gas and lung volume increases their intrapulmonary pressure – the pressure within alveoli – falls
- If lung volume decreases intrapulmonary pressure rises
- If air should flow into lungs, intrapulmonary pressure must be lower than atmospheric pressure;
- Likewise, intrapulmonary pressure above the atmospheric pressure must flow air out of lungs
- These changes – mechanically accomplished by skeletal muscles of thoracic and abdominal walls – changes the volume of thoracic cavity.

Pressure and Flow

- Air flow primarily based on the difference between atmospheric pressure and intrapulmonary pressure
- As atmospheric pressure varies for place and time; pressure is usually referred to as ‘relative’

PRESSURES AT REST

- Atmospheric pressure: 760 mm Hg
- Intra-alveolar pressure: 760 mm Hg
- Intrapleural pressure: 756 mm Hg

ALVEOLAR PRESSURE

- **Alveolar pressure during inspiration:** The alveolar pressure is lower than the atmospheric pressure during inspiration (760 to 758 mm Hg) so air moves from the atmosphere towards the lung alveoli.
- **Alveolar pressure during expiration:** Alveolar pressure is increased during the process of expiration and is slightly more with respect to the atmospheric pressure (763 mm Hg) so air moves from lungs to the atmosphere.

PLEURAL AND TRANSPLEURAL PRESSURE

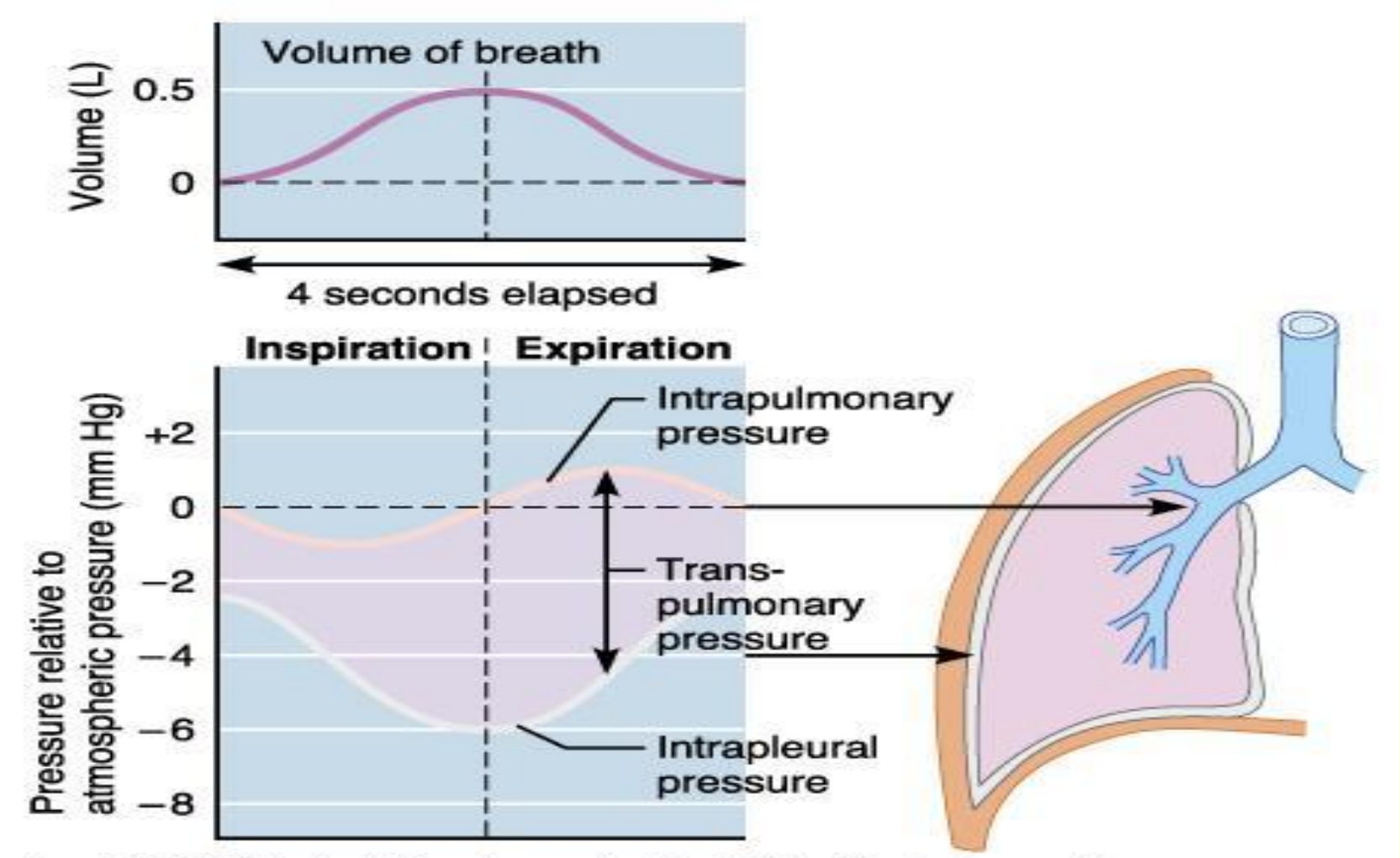
- **Intrapleural pressure or pleural pressure:** Pressure in the narrow space between the lung pleura and chest wall pleura. Negative pressure in the pleural space prevent the collapse of the lungs.

  -5 cm H_2O at the beginning of inspiration

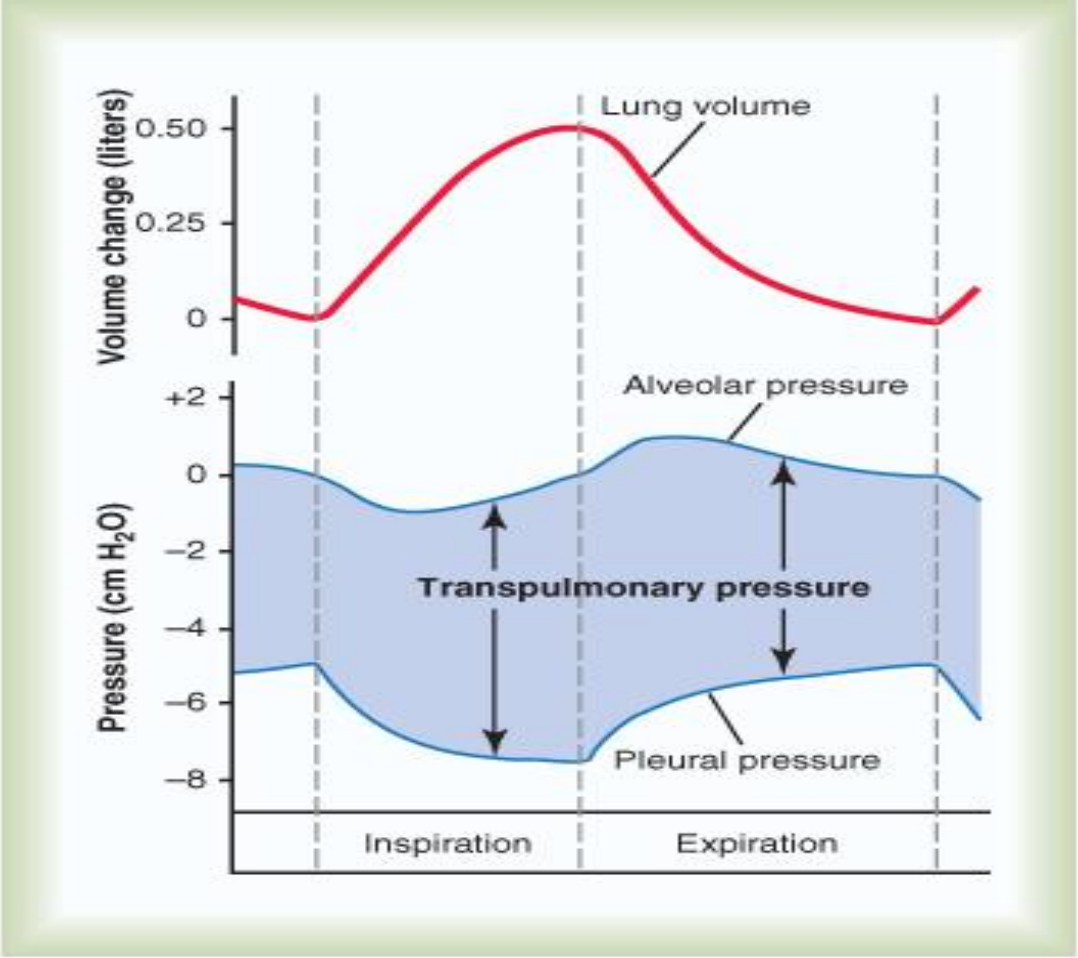
  -7.5 cm H_2O at the peak of inspiration

- **Transpulmonary pressure:** Pressure difference between the alveolar and pleural pressure is called transpulmonary pressure.

CHANGING ALVEOLAR VOLUME



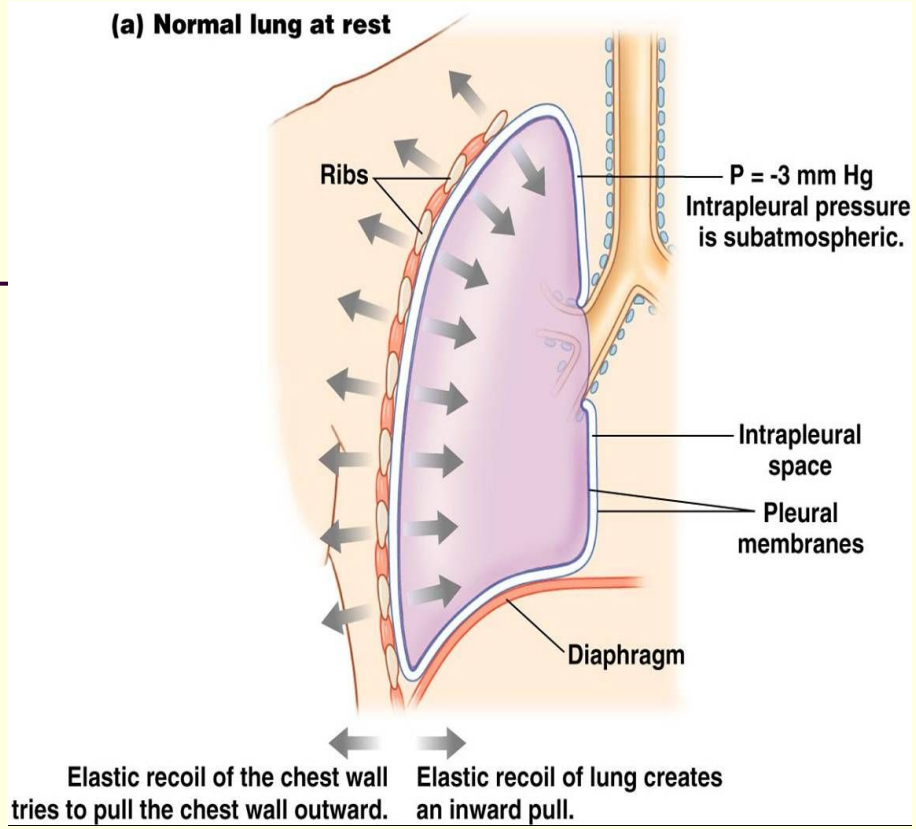
NORMAL BREATHING CYCLE



ROLE OF NEGATIVE INTRAPLEURAL PRESSURE

- At rest the intra pleural pressure is negative (Sub-atmospheric) More negative at apex (-10 cms of water) as compared to base (-2.5 cms of water)
- It can become more negative during deep Inspiration and becomes substantially more positive during Forced expiration
- Main function is that it prevents the lungs from collapsing and the chest wall from going out.

WHY THE INTRAPLEURAL PRESSURE IS SUBATMOSPHERIC?

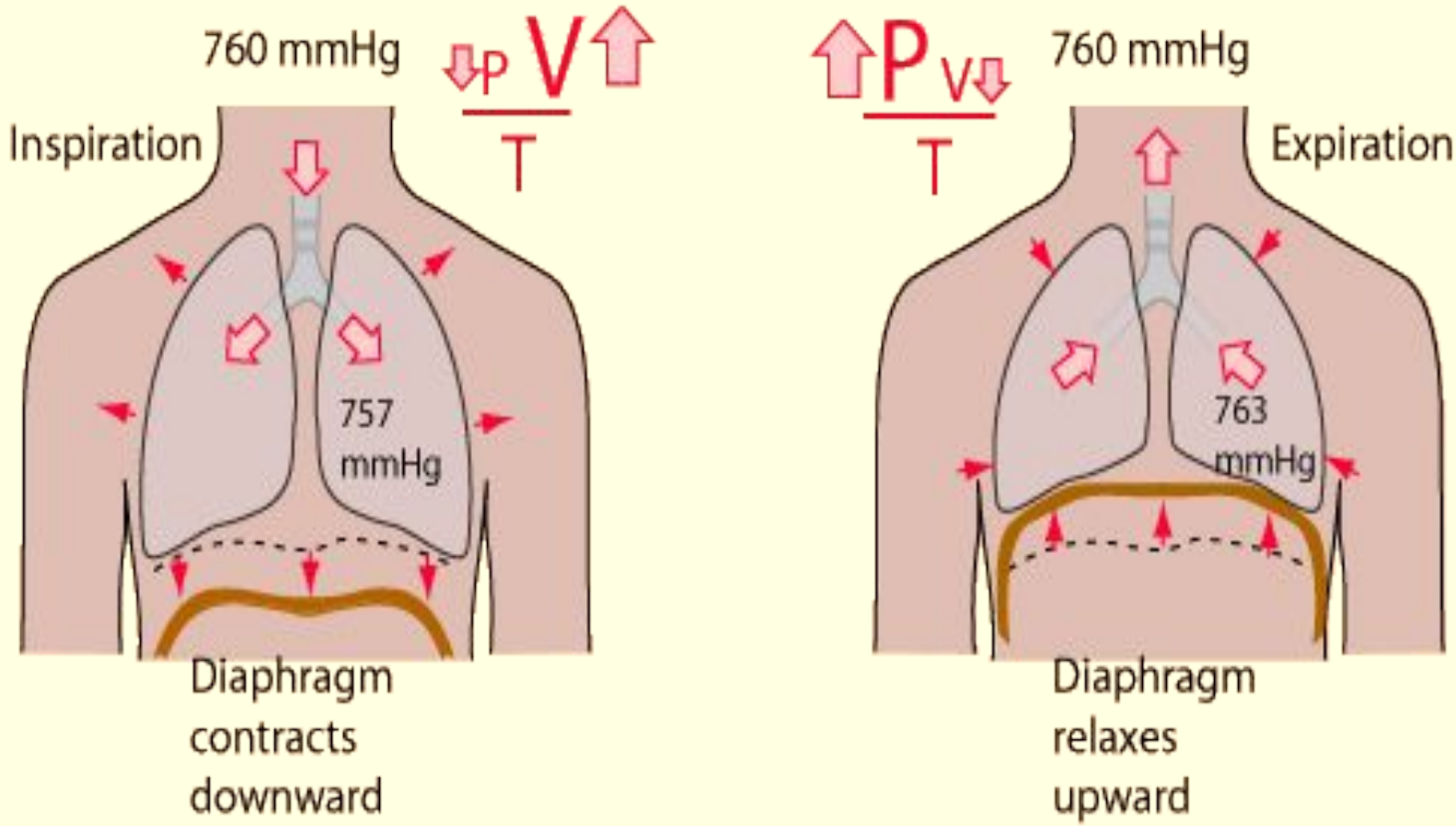


- Because of the lungs' elasticity, they try to pull inward away from the thoracic wall after expansion.
- Likewise, the thoracic wall tends to move outward away from lungs.
- The transmural pressure, prevent these structures from pulling away from each other except to the slightest degree.
- The resultant slight expansion of the pleural cavity is sufficient to drop the pressure in this cavity by 4 mm Hg, bringing the intrapleural pressure to the subatmospheric level of 756(-4) mm Hg.
- This pressure drop occurs because the pleural cavity is filled with fluid, which cannot expand to fill the slightly larger volume.
- Therefore, a vacuum exists in the slightly expanded pleural cavity not occupied by intrapleural fluid, producing a small drop in intrapleural pressure below atmospheric pressure

PULMONARY VENTILATION

- **Pulmonary ventilation** (Pulmo=lungs, ventilation=breathing)
- Pulmonary ventilation is the inspiration (inflow) and expiration (outflow) of air between the atmosphere and lungs
- In ventilation important factor called the pressure gradient exists.
- Air moves into the lungs when the pressure inside the lungs is less than that of the atmospheric pressure.
- Air moves from the lungs to the atmosphere, when the pressure in the lungs is greater than the atmospheric pressure.

PULMONARY VENTILATION [INSPIRATION AND EXPIRATION]



MOVEMENTS OF THORACIC CAGE

- Inspiration causes enlargement of thoracic cage. Thoracic cage enlarges because of increase in all diameters, viz. anteroposterior, transverse and vertical diameters.
- Anteroposterior and transverse diameters of thoracic cage are increased by the elevation of ribs.
- Vertical diameter is increased by the descent of diaphragm. In general, change in the size of thoracic cavity occurs because of the movements of four units of structures:

1. Thoracic lid **2**. Upper costal series

3. Lower costal series **4**. Diaphragm.

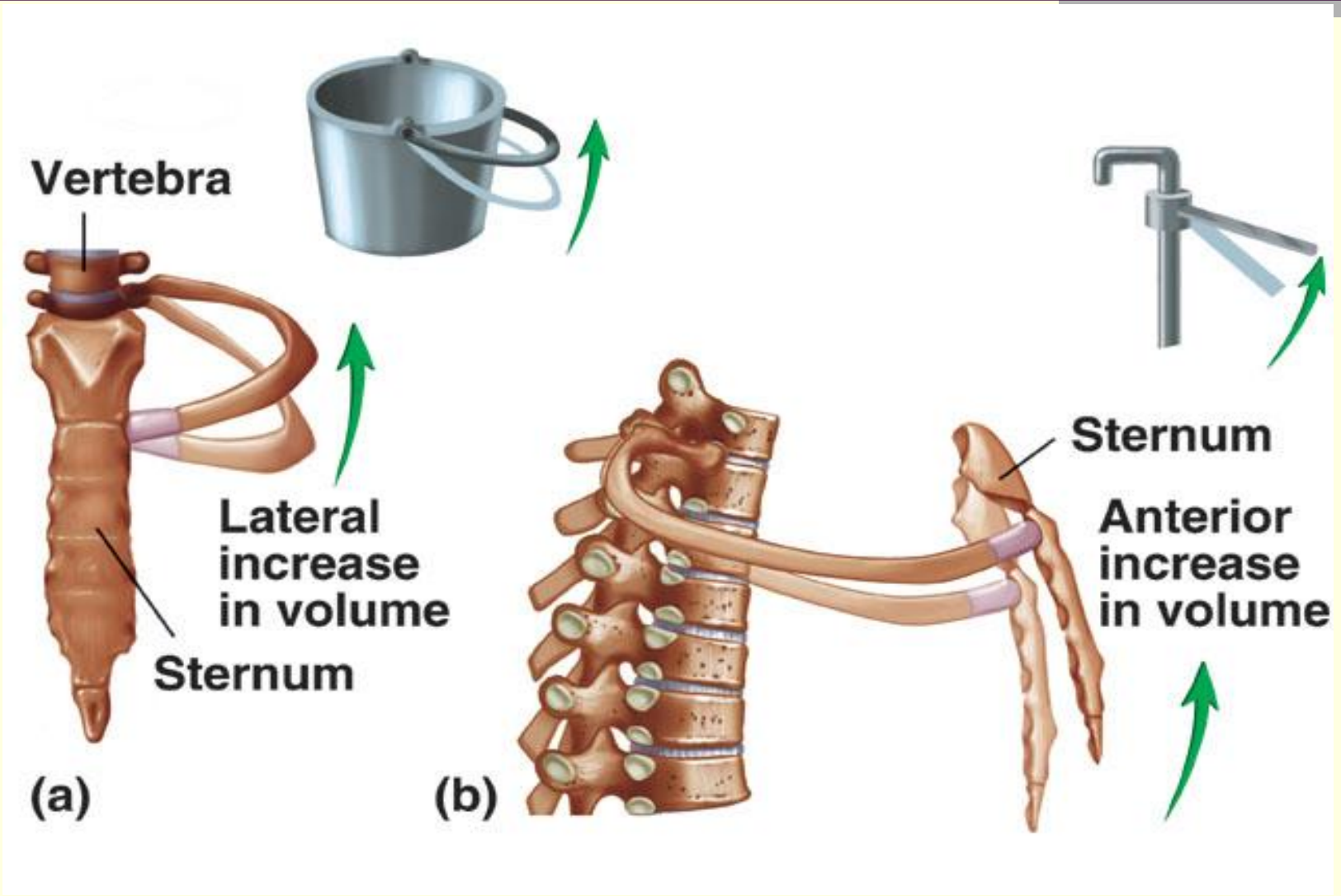
1. Thoracic Lid

- Thoracic lid is formed by manubrium sterni and the first pair of ribs. It is also called thoracic operculum. Movement of thoracic lid increases the anteroposterior diameter of thoracic cage. Due to the contraction of scalene muscles, the first ribs move upwards to a more horizontal position. This increases the anteroposterior diameter of upper thoracic cage.

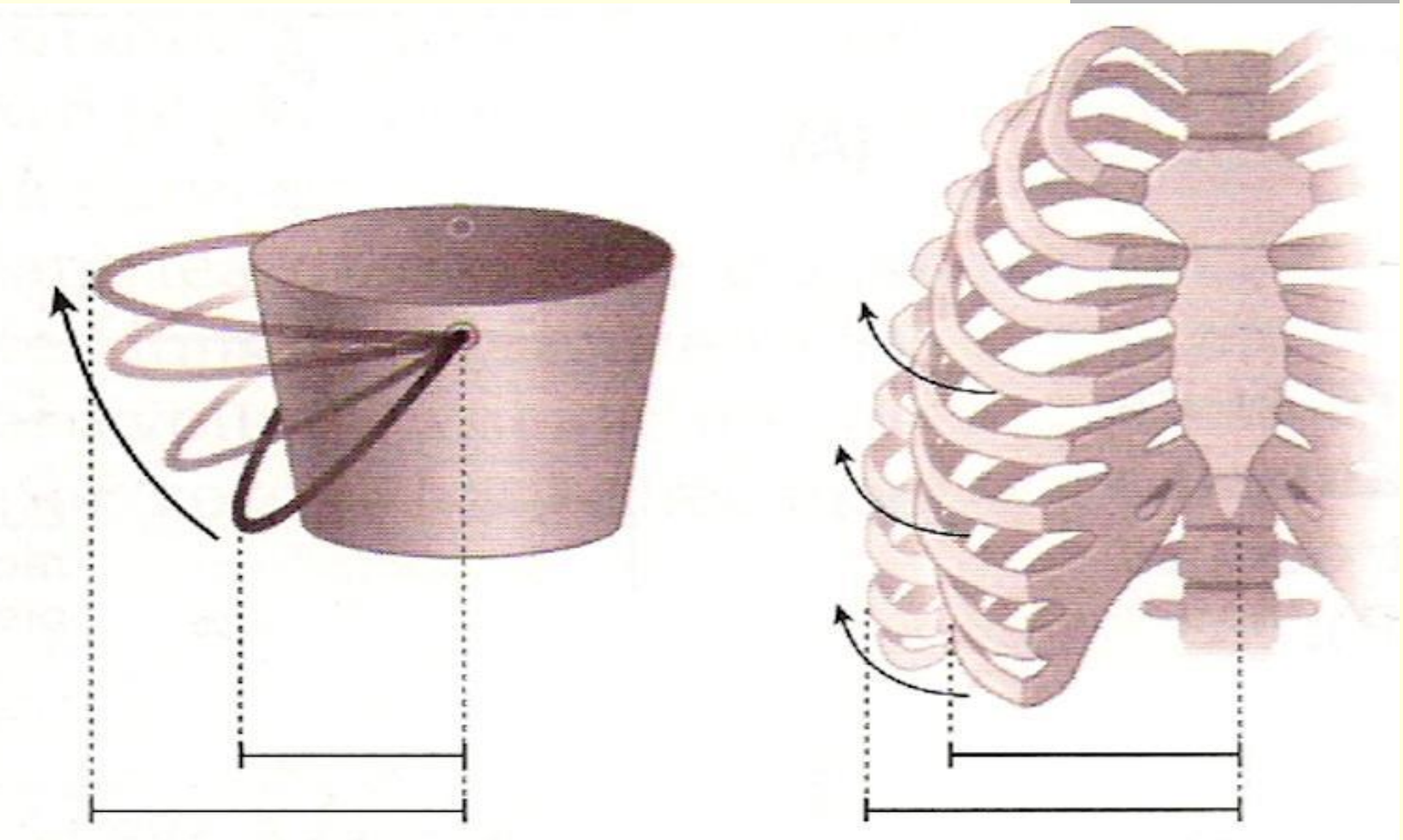
2. Upper Costal Series

- Upper costal series is constituted by second to sixth pair of ribs. Movement of upper costal series increases the anteroposterior and transverse diameter of the thoracic cage. Movement of upper costal series is of two types:
- i. Pump handle movement
- ii. Bucket handle movement.

THORACIC VOLUME



THORACIC VOLUME



3. Lower Costal Series

- Lower costal series includes seventh to tenth pair of ribs. Movement of lower costal series increases the transverse diameter of thoracic cage by bucket handle movement. **Bucket handle movement** Lower costal series of ribs also show bucket handle movement by swinging outward and upward. This movement increases the transverse diameter of the thoracic cage. Eleventh and twelfth pairs of ribs are the floating ribs. These ribs are not involved in changing the size of thoracic cage.

4. Diaphragm

- Movement of diaphragm increases the vertical diameter of thoracic cage. Normally, before inspiration the diaphragm is dome shaped with convexity facing upwards. During inspiration, due to the contraction, muscle fibers are shortened. But the central tendinous portion is drawn downwards so the diaphragm is flattened. Flattening of diaphragm increases the vertical diameter of the thoracic cage.

THORACIC VOLUME

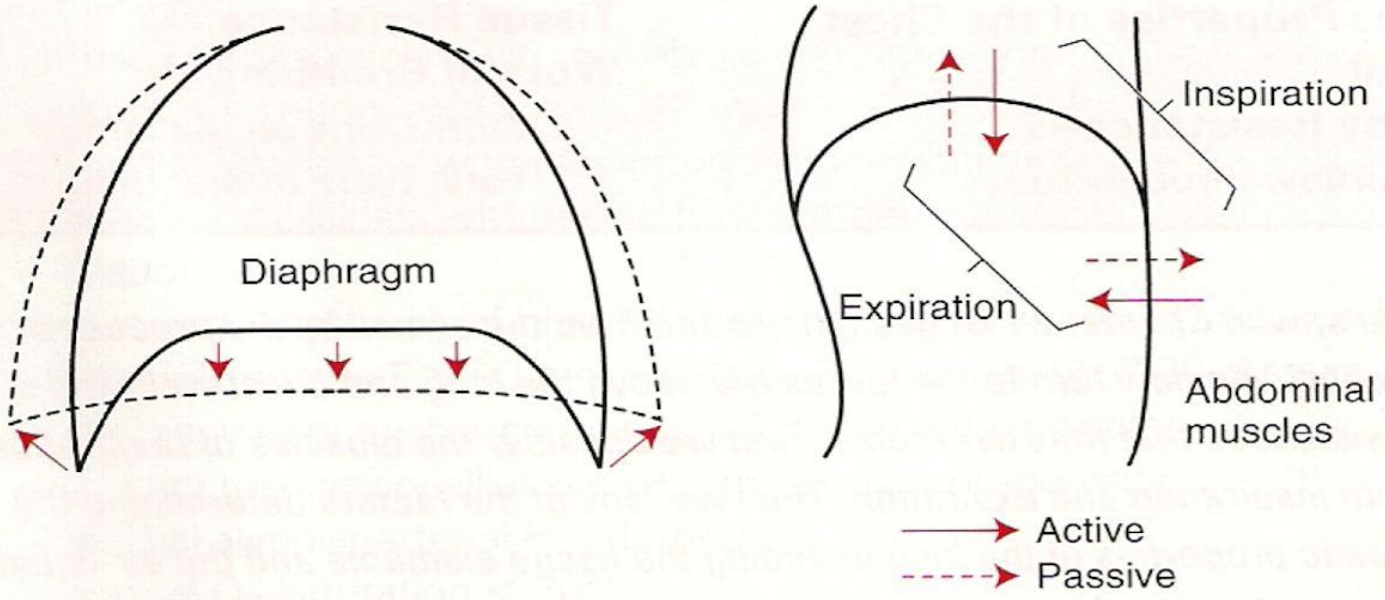


Figure 7-1. On inspiration, the dome-shaped diaphragm contracts, the abdominal contents are forced down and forward, and the rib cage is widened. Both increase the volume of the thorax. On forced expiration, the abdominal muscles contract and push the diaphragm up.

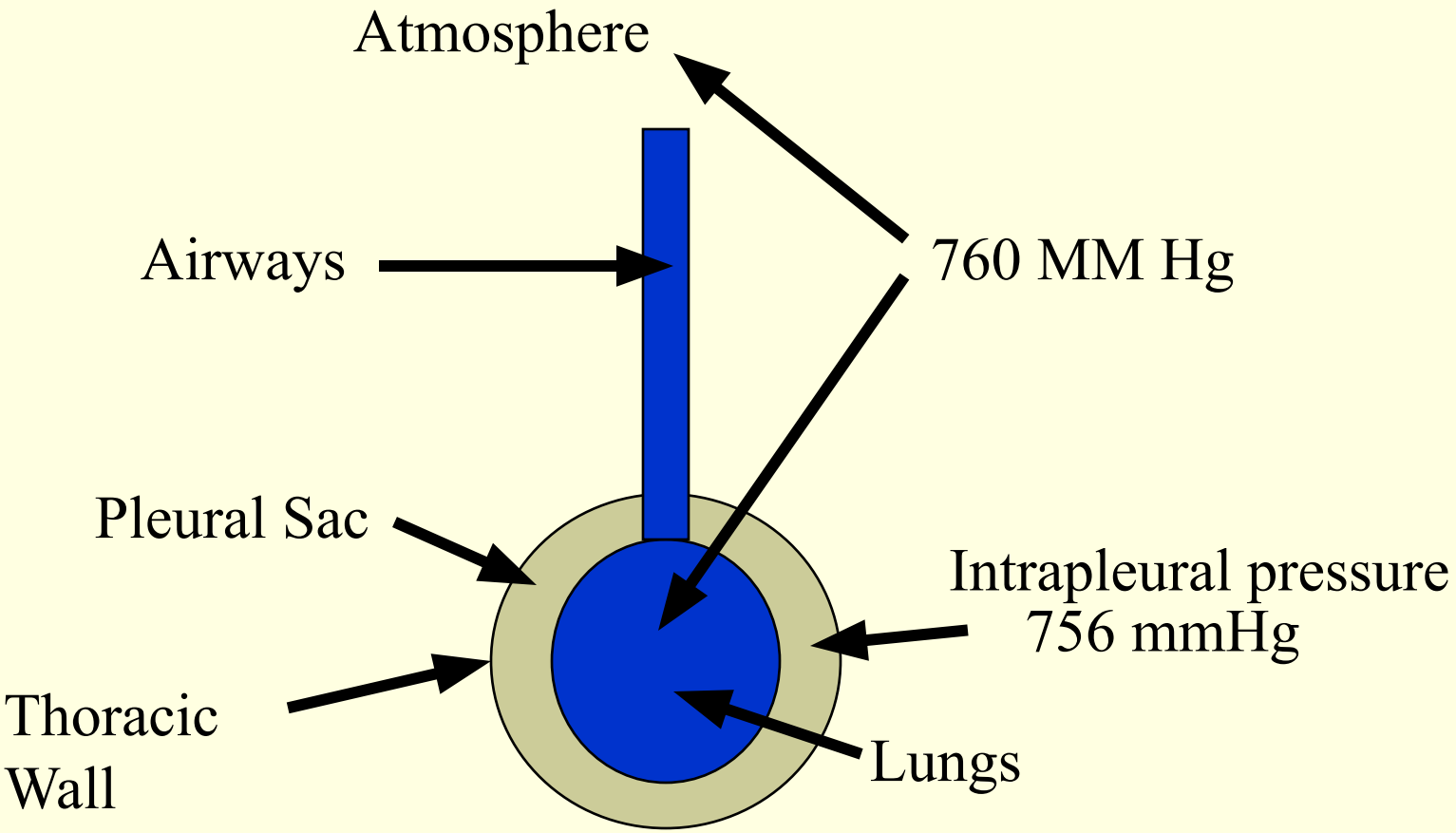
MOVEMENTS OF LUNGS

- During inspiration, due to the enlargement of thoracic cage, the negative pressure is increased in the thoracic cavity. It causes expansion of the lungs. During expiration, the thoracic cavity decreases in size to the pre inspiratory position. Pressure in the thoracic cage also comes back to the pre inspiratory level. It compresses the lung tissues so that, the air is expelled out of lungs.

THE TRANSMURAL PRESSURE GRADIENT INFLATES THE LUNGS

- Thoracic cavity larger than lungs
- Transmural (Across Lung Wall) pressure gradient holds thoracic wall and lungs in close apposition
- This pressure gradient is balanced by the elastic forces in the alveoli producing equilibrium

REST PRESSURES



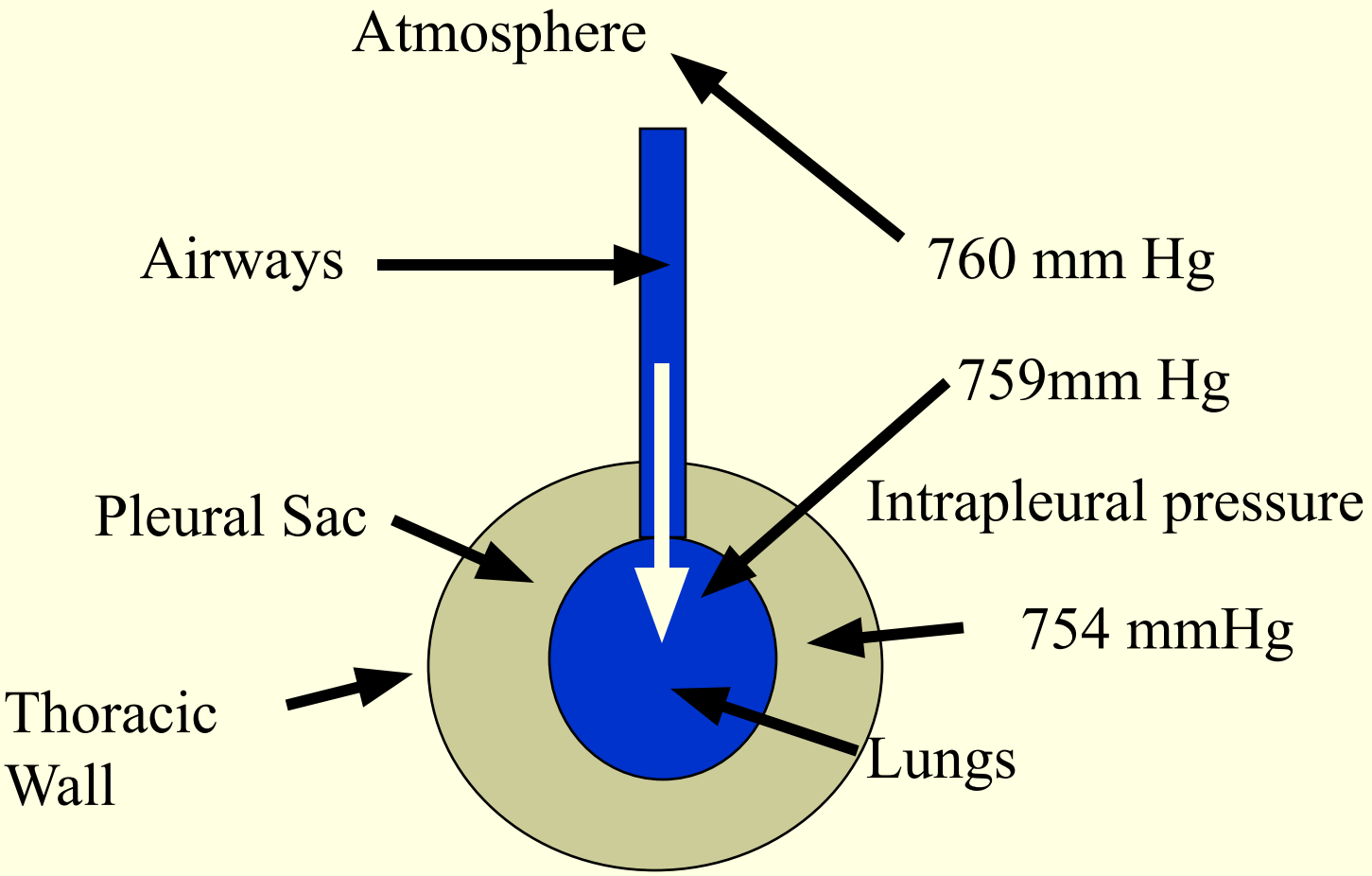
AIR IS A COMPRESSIBLE GAS WHICH OBEYS BOYLE'S LAW

- $P_1V_1 = P_2V_2$
- If Volume increases, Pressure must decrease
- As lungs expand, pressure inside falls

INSPIRATION

- Elevation of ribs expands lungs
- Lowering of diaphragm by contraction also expands lungs
- Expansion of lungs causes pressure inside to drop below atmospheric pressure
- Air rushes in to fill the expanded lungs

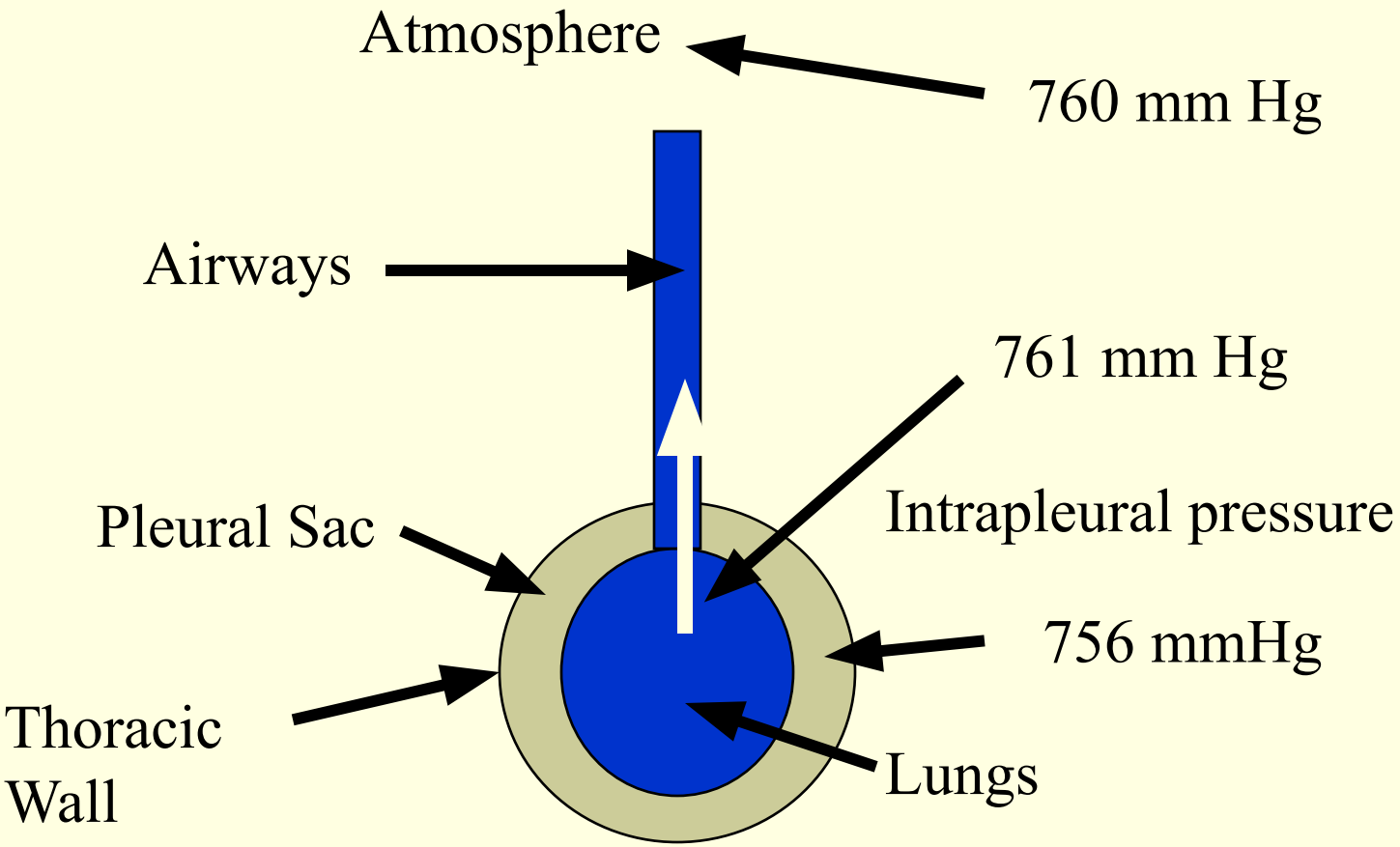
INSPIRATION



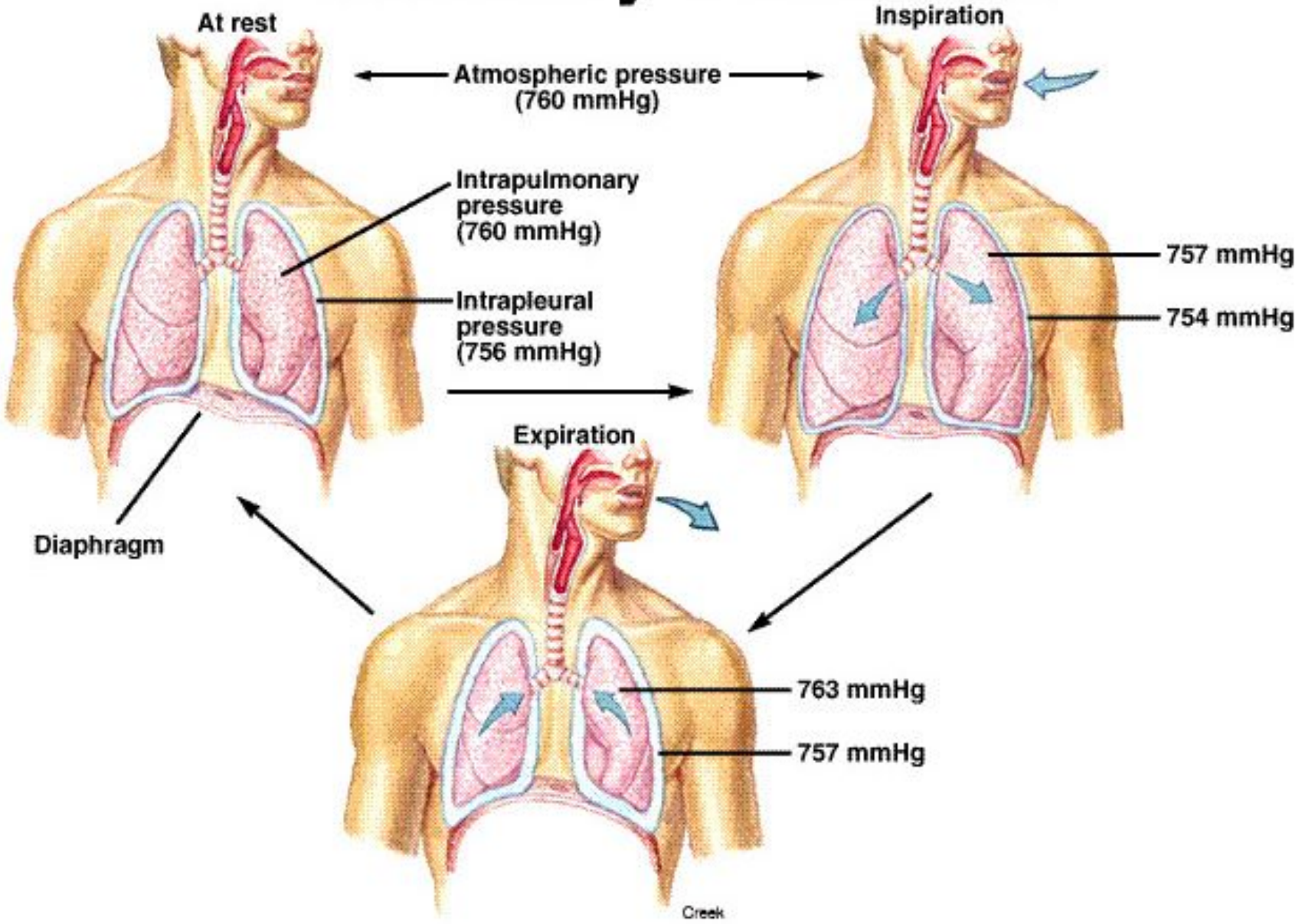
EXPIRATION

- Return of ribs to rest position causes diminishing of lung volume
- Return of diaphragm to rest position also causes diminishing of lung volume
- Diminishing of lung volume causes pressure in lung to raise to a higher value than atmospheric pressure
- Air flows out of the lungs

EXPIRATION


Atmosphere
760 mm Hg
Airways
761 mm Hg
Pleural Sac
Intrapleural pressure
756 mmHg
Thoracic Wall
Lungs

Pulmonary Ventilation



Physical Aspects of Ventilation

- Ventilation occurs as a result in pressure differences induced by changes in lung volume.
- Physical properties that affect lung function:
  - Compliance.
  - Elasticity.
  - Surface Tension.

LUNG VOLUMES

- Tidal Volume (TV): 500 ml
- Inspiratory reserve volume (IRV): 3 liters
- Inspiratory capacity (IC): 3.5 liters
- Expiratory reserve volume (ERV): 1 liter
- Residual volume (RV): 1.2 liters
- Functional Residual Capacity (FRC): 2.2 l
- Vital Capacity (VC): 4.5 liters
- Total Lung Capacity (TLC): 5.7 liters

THANK YOU